USN											

"24CS330"

JSS MAHAVIDYAPEETHA
JSS SCIENCE AND TECHNOLOGY UNIVERSITY, MYSURU

Department of Computer Science and Engineering

III Semester: Test I

Data Structures

Duration: 1hour 15 mins Date: 11-11-2025 Max. Marks: 30 Marks

NOTE: Question 1 is compulsory. Answer any one between question 2 and question 3.

Q.NO	CO	CD	PI	QUESTION	MARKS
1.	CO1	L3	1.7.1	a) Write a program to create an array of integers and perform the following operations by writing function for each operation which takes pointer to array as parameter. i)READ: To read the array elements ii) WRITE: To Display the array elements iii) REVERSE: To reverse the array contents. iv) SEARCHDELETE: To Search for a key value and delete if exists, otherwise a suitable message. **(Create a pointer to this function and call this function using pointer). Assume each value exists only once.**	10
		L2	2.1.2	b) Demonstrate the difference between structures and unions with suitable example.	05
2.	CO1	L3	1.7.1	a) Develop a C program to implement an array of integers and perform the following: i Read the array elements ii Display the array elements iii Perform bubble sort. Note: Illustrate the use of pointer to an array variable and passing and returning of array type to and from the function (by reference). Use dynamic memory allocation mechanism.	10
		L2	2.1.2	b) Illustrate with an example, the problems of not deallocating the dynamically allocated block of memory using free function.	05
				OR	
3.	CO1	L3	1.7.1	a) Write a function called Remove(P,N) where P is a pointer to one dimensional integer array and N is the number of elements in the array. This function should remove all negative elements in the array using P. A suitable message to be displayed if all elements are	08

				positive.	
				b) Write a function in C to find the frequency of the character using the prototype below: int Frequency_char(char str, char ch); where str is the input string and ch is the character whose frequency is to be determined.	07

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation.	L3
CO2	Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list.	L3
CO3	Solve the problems of conversion and evaluation of different types of expressions using stack data structure, comprehend the implementation of different types of Queues.	L3
CO4	Using tree data structures solve the problems of searching, sorting, expression conversion and expression evaluation.	L3
CO5	Applying graph traversal techniques solve the problems of testing connectivity and acyclicity in a graph.	L3

PI's	
1.4.1	Apply theory and principles of computer science and engineering to solve an engineering problem
1.7.1	Apply theory and principles of computer science and engineering to solve an engineering problem.
2.1.2	Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem

Blooms Taxonomy	
L1	Remembering
L2	Understanding
L3	Applying

— End —